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CHAPTER 3

General Vector Spaces

(Part 3)

Coordinates and Change of Basis

Coordinates

Let $B = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots \boldsymbol{v}_n\}$ be an ordered basis for the vector space V. Let $\mathbf{v}$ be a vector in V, and let $c_1, c_2, \ldots c_n$ be a unique scalars such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n$$

Then $c_1, c_2, \ldots c_n$ are called the **coordinates of $\mathbf{v}$ relative to** B, and written as

$$[\boldsymbol{v}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \cdot \\ c_n \end{bmatrix}$$

Example # 1

Let $V = \Re^2$ and B be the ordered basis,

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

Find the coordinates of the vector, $\mathbf{v} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ relative to B.

Solution

The coordinates c_1 and c_2 are found by writing **v** as a linear combination of the two vectors in *B*.

$$c_1\begin{bmatrix}1\\1\end{bmatrix}+c_2\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}1\\5\end{bmatrix}$$

and we obtain $\mathbf{C}_1 = 3$ and $\mathbf{C}_2 = 2$. Therefore the coordinate vector of vector, $\mathbf{v}=\begin{bmatrix}1\\5\end{bmatrix}$ is $[\mathbf{v}]_B=\begin{bmatrix}3\\2\end{bmatrix}$

Example # 2

Let *W* be the subspace of all symmetric matrices in the vector space M_{2x2} . Let

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Find the coordinates of

vector, $\mathbf{v} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ relative to *B.*

Solution

Observe that **v** can be written as

$$2\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}+3\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}+5\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}=\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix}$$

Then relative to the ordered basis *B*, the coordinate vector of **v** is

$$[v]_B=\begin{bmatrix}2\\ 3\\ 5\end{bmatrix}$$

Change of Basis

Change from coordinates relative to one basis for V to another basis for V.

Example

Let *V* be a vector space of dimension 2 and let $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $B' = \{\mathbf{v}'_1, \mathbf{v}'_2\}$ be ordered bases for *V*. Now let **v** be a vector in *V,* and suppose that the coordinates of **v** relative to *B* are given by

$$[\mathbf{v}]_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{that is} \quad \mathbf{v} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2$$

Solution

- To determine the coordinates of **v** relative to *B*'

1) Write $\mathbf{v}_1$ and $\mathbf{v}_2$ in terms of the vectors $\mathbf{v}'_1$ and $\mathbf{v}'_2$

v_2

.

$$\mathbf{v}_1 = a_1\mathbf{v}'_1 + a_2\mathbf{v}'_2$$

$$\mathbf{v}_2 = b_1\mathbf{v}'_1 + b_2\mathbf{v}'_2$$

Since *B*' is a basis, there are scalars a_1, a_2, b_1 and b_2

Solution (cont.)

2) Then **v** can be write as

$$\mathbf{v} = x_1(a_1\mathbf{v}'_1 + a_2\mathbf{v}'_2) + x_2(b_1\mathbf{v}'_1 + b_2\mathbf{v}'_2)$$

v_2

3) Collecting the coefficient s of $\mathbf{v}'_1$ and $\mathbf{v}'_2$ gives

$$\mathbf{v} = (x_1a_1 + x_2b_1)\mathbf{v}'_1 + (x_1a_2 + x_2b_2)\mathbf{v}'_2$$

Solution (cont.)

4) Coordinates of **v** relative to the basis *B*' are given by

$$[\mathbf{v}]_{B'} = \begin{bmatrix} x_1 a_1 + x_2 b_1 \\ x_1 a_2 + x_2 b_2 \end{bmatrix}$$

5) Rewriting the vector on the right-hand side as a matrix product, we have

$$[v]_{B'} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} [v]_B$$

Solution (cont.)

6) Notice that the column vectors of the matrix are the coordinate vectors $[\mathbf{v}_1]_{B'}$ **and** $[\mathbf{v}_2]_{B'}$

The matrix $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ is called the **transition matrix**

from *B* to *B*' and is denote by $[I]_B^{B'}$, so that

$$[\mathbf{v}]_{B'} = [I]_B^{B'}[\mathbf{v}]_B$$

Example

- Let $V = \Re^2$ with bases

$$\boldsymbol{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \text{ and } B' = \left\{ \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

a. Find the transition matrix from *B* to *B'*.

b. Let $[\mathbf{v}]_{\boldsymbol{B}} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ **and find** $[\mathbf{v}]_{\boldsymbol{B}'}$

Solution

a. By denoting the vectors in B by $\mathbf{v}_1$ and $\mathbf{v}_2$ and those in B' by $\mathbf{v}'_1$ and $\mathbf{v}'_2$, the column vectors of the transition matrix are $[\mathbf{v}_1]_{B'}$ and $[\mathbf{v}_2]_{B'}$. These coordinate vectors are found from the equations

$$c_1\begin{bmatrix} 2 \\ -1 \end{bmatrix} + c_2\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad d_1\begin{bmatrix} 2 \\ -1 \end{bmatrix} + d_2\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Solving these equations gives $c_1 = 2$ and $c_2 = 3$, and $d_1 = 0$ and $d_2 = -1$, so that

$$[\mathbf{v}_1]_{B'} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \text{and} \quad [\mathbf{v}_2]_{B'} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Therefore, the transition matrix is

$$[I]_B^{B'} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}$$

Solution (cont.)

b. Since

$$[\mathbf{v}]_{B'} = [I]_B^{B'}[\mathbf{v}]_B$$

then

$$[\mathbf{v}]_{B'} = \begin{bmatrix} 2 & 0 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix}$$

Observe that the same vector, relative to the different bases, is obtained from the coordinates $[\mathbf{v}]_B$ and $[\mathbf{v}]_{B'}$. That is,

$$3\begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 6\begin{bmatrix} 2 \\ -1 \end{bmatrix} + 11\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Example

Let $B = \{e_1, e_2\}$ be a standard ordered basis for $\Re^2$

Let *B'* be the ordered basis given by

$$B' = \{v'_1, v'_2\} = \left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$$

a. Find the **transition matrix** from *B* to *B'*.

b. Let $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, find $[v]_{B'}$

Solution

a. The transition matrix from B to B' is computed by solving the equations

$$c_1\begin{bmatrix}-1\\1\end{bmatrix}+c_2\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix} \quad \text{and} \quad d_1\begin{bmatrix}-1\\1\end{bmatrix}+d_2\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}0\\1\end{bmatrix}$$

That is, we must solve the linear systems

$$\begin{cases}-c_1+c_2=1\\ c_1+c_2=0\end{cases} \quad \text{and} \quad \begin{cases}-d_1+d_2=0\\ d_1+d_2=1\end{cases}$$

The unique solutions are given by $c_1=-\frac{1}{2}, c_2=\frac{1}{2}$ and $d_1=\frac{1}{2}, d_2=\frac{1}{2}$. The transition matrix is then given by

$$[I]_B^{B'}=\begin{bmatrix}-\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}$$

Solution (cont.)

b.

$$[\mathbf{v}]_{B'} = [I]_{B}^{B'}[\mathbf{v}]_{B}$$

$$[\mathbf{v}]_{B'} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{7}{2} \end{bmatrix}$$

Exercise #1

Find the coordinates of the vector **v** relative to the ordered basis *B*.

$$a)\ \ \boldsymbol{B} = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\},\ \ \mathbf{v} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}$$

$$b)\ \ \boldsymbol{B} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\},\ \ \mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 9 \end{bmatrix}$$

Exercise #2

i) Find the **transition matrix** between the ordered

bases $\boldsymbol{B}_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ **and** $\boldsymbol{B}_2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

ii) Given $[\mathbf{v}]_{\boldsymbol{B}_1} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, **find** $[\mathbf{v}]_{\boldsymbol{B}_2}$